

P452 Project 2: Simulation of Many-Body System

Streamlit Website Link: <https://quantumgasesopticschengchincourse-5zau9dzwe8ovrd6p8ynjhs.streamlit.app/Phase>

1: Exact Diagonalization of the Heisenberg Model 1.1: The 2-Site Dimer and Spin Frustration (Pen and Paper) 1. Hilbert Space For $N=2$ spins with $S=1/2$, each site has 2 states (up or down). The total dimension of the Hilbert space is 4. The basis states are: $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$.

2: Hamiltonian Matrix Expansion The Hamiltonian is given by

$\hat{H} = J(\hat{S}_1 \cdot \hat{S}_2) + H(\hat{S}_1^z + \hat{S}_2^z)$. We expand the interaction term using ladder operators:

$$\hat{S}_1 \cdot \hat{S}_2 = \hat{S}_1^z \hat{S}_2^z + \frac{1}{2}(\hat{S}_1^+ \hat{S}_2^- + \hat{S}_1^- \hat{S}_2^+)$$

Now, let's apply this Hamiltonian step-by-step to our basis states: For $|\uparrow\uparrow\rangle$: The ladder operators yield zero because the spins cannot be raised further.

$$\hat{S}_1^z \hat{S}_2^z |\uparrow\uparrow\rangle = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) |\uparrow\uparrow\rangle = \frac{1}{4} |\uparrow\uparrow\rangle$$

$$\hat{H}_{\text{Zeeman}} |\uparrow\uparrow\rangle = H \left(\frac{1}{2} + \frac{1}{2}\right) |\uparrow\uparrow\rangle = H |\uparrow\uparrow\rangle$$

$$\hat{H} |\uparrow\uparrow\rangle = \left(\frac{J}{4} + H\right) |\uparrow\uparrow\rangle$$

For $|\downarrow\downarrow\rangle$: Similarly, ladder operators yield zero.

$$\hat{S}_1^z \hat{S}_2^z |\downarrow\downarrow\rangle = \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) |\downarrow\downarrow\rangle = \frac{1}{4} |\downarrow\downarrow\rangle$$

$$\hat{H}_{\text{Zeeman}} |\downarrow\downarrow\rangle = H \left(-\frac{1}{2} - \frac{1}{2}\right) |\downarrow\downarrow\rangle = -H |\downarrow\downarrow\rangle$$

$$\hat{H} |\downarrow\downarrow\rangle = \left(\frac{J}{4} - H\right) |\downarrow\downarrow\rangle$$

For $|\uparrow\downarrow\rangle$: The $\hat{S}_1^z \hat{S}_2^z$ term gives $-1/4$. The ladder operator $\hat{S}_1^- \hat{S}_2^+$ flips the spins.

$$\frac{J}{2}(\hat{S}_1^+ \hat{S}_2^- + \hat{S}_1^- \hat{S}_2^+) |\uparrow\downarrow\rangle = \frac{J}{2} |\downarrow\uparrow\rangle$$

$$\hat{H}_{\text{Zeeman}} |\uparrow\downarrow\rangle = H \left(\frac{1}{2} - \frac{1}{2}\right) |\uparrow\downarrow\rangle = 0$$

$$\hat{H}|\uparrow\downarrow\rangle = -\frac{J}{4}|\uparrow\downarrow\rangle + \frac{J}{2}|\downarrow\uparrow\rangle$$

For $|\downarrow\uparrow\rangle$: By the same logic as above:

$$\hat{H}|\downarrow\uparrow\rangle = -\frac{J}{4}|\downarrow\uparrow\rangle + \frac{J}{2}|\uparrow\downarrow\rangle$$

This gives the matrix in the ordered basis ($|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$):

$$H = \begin{pmatrix} \frac{J}{4} + H & 0 & 0 & 0 \\ 0 & -\frac{J}{4} & \frac{J}{2} & 0 \\ 0 & \frac{J}{2} & -\frac{J}{4} & 0 \\ 0 & 0 & 0 & \frac{J}{4} - H \end{pmatrix}$$

3: Ground State Transition To find the energies of the mixed states, we solve the eigenvalue problem for the inner 2×2 block:

$$\det \begin{pmatrix} -\frac{J}{4} - \lambda & \frac{J}{2} \\ \frac{J}{2} & -\frac{J}{4} - \lambda \end{pmatrix} = 0$$

$$\left(-\frac{J}{4} - \lambda\right)^2 - \left(\frac{J}{2}\right)^2 = 0 \implies -\frac{J}{4} - \lambda = \pm \frac{J}{2}$$

$$\lambda = -\frac{J}{4} \pm \frac{J}{2}$$

This yields the Singlet $|S\rangle$ and Triplet $|T_0\rangle$ state energies alongside the polarized states:

$E_{T+} = \frac{J}{4} + H$ (from $|\uparrow\uparrow\rangle$) $E_{T-} = \frac{J}{4} - H$ (from $|\downarrow\downarrow\rangle$) $E_S = -\frac{3J}{4}$ (Singlet: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$) $E_{T0} = \frac{J}{4}$ (Triplet 0: $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$) At $H=0$, the ground state is the Singlet ($E = -3J/4$), representing an anti-ferromagnetic configuration. Assuming $H < 0$ (such that the field points "up" and lowers the energy of $|\uparrow\uparrow\rangle$), the critical field H_c occurs when the fully polarized state crosses the singlet:

$$\frac{J}{4} - |H_c| = -\frac{3J}{4} \implies |H_c| = J$$

4: Spin Frustration (3 Spins on a Ring) Symmetry: The total Hilbert space dimension is 8.

Because $[\hat{H}, \hat{S}_{\text{total}}^z] = 0$, the Hamiltonian cannot flip the total number of up/down spins.

The matrix block-diagonalizes into sectors: $S^z = 3/2$ (1 state), $1/2$ (3 states), $-1/2$ (3 states), and $-3/2$ (1 state). Calculation: * Sector $+3/2$ ($|\uparrow\uparrow\uparrow\rangle$): $E = \frac{3J}{4} + \frac{3H}{2}$. Sector $+1/2$ (Basis: $|\uparrow\uparrow\downarrow\rangle, |\uparrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\rangle$): Evaluating the diagonal elements for $|\uparrow\uparrow\downarrow\rangle$, we have two favorable Anti-Ferromagnetic (AF) bonds (up-down, down-up) contributing $-J/4 - J/4$ and one unfavorable Ferromagnetic (FM) bond (up-up) contributing $+J/4$. Total diagonal J energy $= -J/4$. The Zeeman energy is $H(1/2 + 1/2 - 1/2) = H/2$. The off-diagonal ladder operators flip anti-aligned neighbors with amplitude $J/2$. The block is:

$$H_{1/2} = \begin{pmatrix} -\frac{J}{4} + \frac{H}{2} & \frac{J}{2} & \frac{J}{2} \\ \frac{J}{2} & -\frac{J}{4} + \frac{H}{2} & \frac{J}{2} \\ \frac{J}{2} & \frac{J}{2} & -\frac{J}{4} + \frac{H}{2} \end{pmatrix}$$

To diagonalize this matrix, let $x = -\frac{J}{4} + \frac{H}{2}$ and $y = \frac{J}{2}$. This is a matrix of the form $(x - y)I + yJ_3$ (where J_3 is a 3×3 matrix of ones). The eigenvalues of J_3 are 3 and 0 (double). Therefore, the eigenvalues are $\lambda_1 = x + 2y$ and $\lambda_{2,3} = x - y$. Substituting x and y back in: $\lambda_1 = -\frac{J}{4} + \frac{H}{2} + 2\left(\frac{J}{2}\right) = \frac{3J}{4} + \frac{H}{2}$ (non-degenerate)
 $\lambda_{2,3} = -\frac{J}{4} + \frac{H}{2} - \frac{J}{2} = -\frac{3J}{4} + \frac{H}{2}$ (doubly degenerate) Analysis: At $H=0$, the ground state is in the $\pm 1/2$ sectors with $E = -3J/4$. It is 4-fold degenerate (two states from $+1/2$, two from $-1/2$). At high $H \gg J$, the fully polarized state becomes the unique ground state. The 4-fold degeneracy at $H=0$ physically represents frustration: on a triangle, you cannot anti-align all three spins simultaneously. If spin 1 is UP and spin 2 is DOWN, spin 3 must geometrically align with one of them, paying an energy penalty. Checkpoint 1.2: From 2x2 to 6x6 Square Lattices 1. Hilbert Space and Dimension For a 2×2 lattice, $N=4$. Total dimension = 16. For the $S_{\text{total}}^z = 0$ sector (2 up, 2 down), the dimension is derived from the combination $D = \binom{4}{2} = \frac{4!}{2!2!} = 6$. The basis states are: $|\uparrow\uparrow\downarrow\downarrow\rangle, |\uparrow\downarrow\uparrow\downarrow\rangle, |\uparrow\downarrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\downarrow\rangle, |\downarrow\uparrow\downarrow\uparrow\rangle, |\downarrow\downarrow\uparrow\uparrow\rangle$. 2. Role of the External Field The Zeeman term is $\hat{H}_{\text{Zeeman}} = H \sum \hat{S}_i^z$. Since the operator $\sum \hat{S}_i^z = \hat{S}_{\text{total}}^z$, for any state $|\phi\rangle$ in the $S_{\text{total}}^z = 0$ sector, $\hat{S}_{\text{total}}^z |\phi\rangle = 0$. Therefore, $H \sum \hat{S}_i^z |\phi\rangle = 0$. The external field only shifts the overall energy of sectors with net magnetization; it does not split or shift levels within the zero-magnetization block. 3. Eigenenergies and Limiting Regimes $H=0$ or $J=0$: At $J=0$, spins don't interact. At $H=0$, $J > 0$, the system minimizes energy via anti-ferromagnetic coupling. The ground state lies in the $S_{\text{total}}^z = 0$ sector. A classical Neel state (up-down-up-down) has energy $-4 \times \frac{J}{4} = -J$. However, quantum fluctuations (the $S^+ S^-$ terms) further lower the quantum ground state energy below the classical value. General case $H, J > 0$: As H increases, it tilts the energy of sectors with higher magnetization downwards. The ground state transitions from $S^z = 0 \rightarrow 1 \rightarrow 2$. These critical fields H_c create the "magnetization staircase." In the thermodynamic limit ($N \rightarrow \infty$), the magnetization steps smooth out into a continuous phase transition. The critical value $(H/J)_c$ is defined as the exact field where the macroscopic system reaches full saturation $\langle M_z \rangle = 1$. Phase 2: Hydrodynamics of Bose-Fermi Mixtures Checkpoint 2.1: Uniform Gas in the Ground State 1. Bosonic Chemical Potential In the Thomas-Fermi approximation, we neglect the kinetic term $-\frac{\hbar^2 \nabla^2}{2m_B}$. Without fermions ($a_{BF} = 0$) and trap ($V_B = 0$), the Gross-Pitaevskii equation reduces to:

$$\mu_B = g_B |\Psi_B|^2 = g_B n_B$$

Compressibility is $\kappa \propto \left(\frac{\partial \mu_B}{\partial n_B}\right)^{-1}$. Taking the derivative:

$$\frac{\partial \mu_B}{\partial n_B} = g_B = \frac{4\pi \hbar^2 a_B}{m_B}$$

As long as $a_B > 0$ (repulsive bosons), compressibility is positive, and the BEC is stable against collapse. 2. Fermionic "Interaction" For a uniform Fermi gas,

$\mu_F = E_F = \frac{\hbar^2}{2m_F} (6\pi^2 n_F)^{2/3}$. In the hydrodynamic equation, the term $K_F(n_F)$ must equate to this chemical potential to ensure the system rests at the Fermi energy. Thus:

$$K_F(n_F) = \frac{\hbar^2}{2m_F} (6\pi^2 n_F)^{2/3}$$

Scaling difference: The bosonic mean-field term scales linearly with density (n_B^1), while the fermionic Pauli pressure term scales as $n_F^{2/3}$. 3. Interspecies Coupling With $a_{BF} \neq 0$, the uniform coupled equations become:

$$\mu_B = g_B n_B + g_{BF} n_F$$

$$\mu_F = E_F(n_F) + g_{BF} n_B$$

To find the effective fermionic interaction, we isolate n_B from the bosonic equation:

$$n_B = \frac{\mu_B - g_{BF} n_F}{g_B}$$

Substitute this into the fermionic equation:

$$\mu_F = E_F(n_F) + g_{BF} \left(\frac{\mu_B - g_{BF} n_F}{g_B} \right)$$

$$\mu_F = E_F(n_F) + \frac{g_{BF} \mu_B}{g_B} - \frac{g_{BF}^2}{g_B} n_F$$

This creates an effective fermionic interaction $g_{F,\text{eff}} = -\frac{g_{BF}^2}{g_B}$. Notice it is always negative (attractive) regardless of the sign of g_{BF} . Stability Analysis: The mixture is stable if the determinant of the interaction matrix is positive ($\det(M) > 0$):

$$M = \begin{pmatrix} g_B & g_{BF} \\ g_{BF} & \frac{\partial E_F}{\partial n_F} \end{pmatrix}$$

First, compute the derivative of the Fermi energy with respect to density:

$$\frac{\partial E_F}{\partial n_F} = \frac{\partial}{\partial n_F} \left[\frac{\hbar^2}{2m_F} (6\pi^2)^{2/3} n_F^{2/3} \right] = \frac{\hbar^2}{2m_F} (6\pi^2)^{2/3} \left(\frac{2}{3} \right) n_F^{-1/3} = \frac{\hbar^2}{3m_F} (6\pi^2)^{2/3} n_F^{-1/3}$$

Now, evaluate the determinant:

$$\det(M) = g_B \left(\frac{\hbar^2}{3m_F} (6\pi^2)^{2/3} n_F^{-1/3} \right) - g_{BF}^2 > 0$$

Checkpoint 2.2: Mixtures in a Trap 1 & 2. Thomas-Fermi and LDA Limits From the coupled equations including the harmonic trap potential $V_i(r) = \frac{1}{2} m_i \omega_i^2 r^2$: Bosons:

$$\mu_B = g_B n_B(r) + g_{BF} n_F(r) + V_B(r)$$

$$n_B(r) = \max \left(0, \frac{\mu_B - V_B(r) - g_{BF} n_F(r)}{g_B} \right)$$

Fermions:

$$\mu_F = \frac{\hbar^2}{2m_F} (6\pi^2 n_F(r))^{2/3} + g_{BF} n_B(r) + V_F(r)$$

Solving for $n_F(r)$:

$$n_F(r) = \max \left(0, \frac{1}{6\pi^2} \left[\frac{2m_F}{\hbar^2} (\mu_F - V_F(r) - g_{BF} n_B(r)) \right]^{3/2} \right)$$

3: Physical Intuition Attractive ($g_{BF} < 0$): The $-g_{BF} n_F(r)$ term effectively increases the local chemical potential for bosons where fermions are dense, and vice versa. They mutually attract, pulling each other into a localized, dense core at the trap center. If attraction is too strong, it violates the determinant condition calculated above, leading to mechanical collapse. Repulsive ($g_{BF} > 0$): The species repel each other. If g_{BF} is large enough to violate the stability condition, they undergo phase separation. The denser gas (usually the BEC due to higher local density) sits at the center, acting as a "plunger" that pushes the fermions into an outer shell at the edges of the trap.

```
In [3]: import numpy as np
import scipy.sparse as sp
import scipy.sparse.linalg as sla
import matplotlib.pyplot as plt
from tqdm import tqdm

def get_basis_sz(N, k):
    """Generate basis states (as integers) for N spins with k spins pointing U
    basis = []
    for i in range(2**N):
        if bin(i).count('1') == k:
            basis.append(i)
    return basis

def build_hamiltonian(N, J, H_field, Sz_target=None):
    """Builds the sparse Hamiltonian for an LxL square lattice with PBC."""
    L = int(np.sqrt(N))
    #define bonds for 2D square lattice with periodic boundary conditions
    bonds = []
    for i in range(L):
        for j in range(L):
            idx = i * L + j
            right = i * L + (j + 1) % L
            down = ((i + 1) % L) * L + j
            bonds.extend([(idx, right), (idx, down)])

    if Sz_target is None:
        basis = list(range(2**N))
    else:
        k = int(N/2 + Sz_target) #number of up spins
```

```

        basis = get_basis_sz(N, k)

    dim = len(basis)
    state_to_idx = {state: i for i, state in enumerate(basis)}

    rows, cols, data = [], [], []

    for i, state in enumerate(basis):
        #zeeman term (H * sum S_z)
        sz_total = sum(0.5 if (state & (1 << j)) else -0.5 for j in range(N))
        rows.append(i); cols.append(i); data.append(H_field * sz_total)

        #interaction term J(SzSz + 1/2(S+S- + S-S+))
        for s1, s2 in bonds:
            sz1 = 0.5 if (state & (1 << s1)) else -0.5
            sz2 = 0.5 if (state & (1 << s2)) else -0.5

            #diag J * Sz * Sz
            rows.append(i); cols.append(i); data.append(J * sz1 * sz2)

            #off diagonal (spin flip) if spins are anti aligned
            if sz1 != sz2:
                # Flip bits s1 and s2
                new_state = state ^ ((1 << s1) | (1 << s2))
                if new_state in state_to_idx:
                    j = state_to_idx[new_state]
                    rows.append(i); cols.append(j); data.append(J / 2.0)

    return sp.csr_matrix((data, (rows, cols)), shape=(dim, dim))

#compute for N=4, 16
N_list = [4, 16]
J = 1.0
H_vals = np.linspace(0, 10, 50)

plt.figure(figsize=(10, 4))

for N in N_list:
    magnetizations = []
    for H in H_vals:
        min_E = float('inf')
        gs_mag = 0
        #iterate over all symmetry sectors to find global ground state
        for Sz in tqdm(range(-N//2, N//2 + 1)):
            H_mat = build_hamiltonian(N, J, H, Sz_target=Sz)

            if H_mat.shape[0] <= 2:
                evals = H_mat.toarray().diagonal()
            else:
                evals = sla.eigsh(H_mat, k=1, which='SA', return_eigenvectors=False)

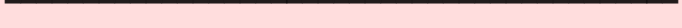
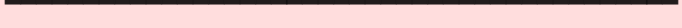
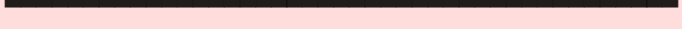
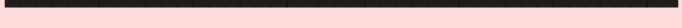
            E0 = np.min(evals)
            if E0 < min_E:
                min_E = E0
                gs_mag = Sz / (N/2) #norm magnetization (-1 to 1)

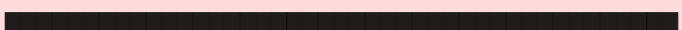
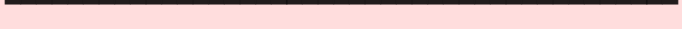
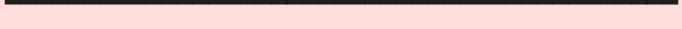
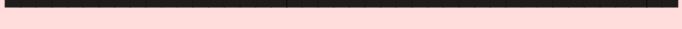
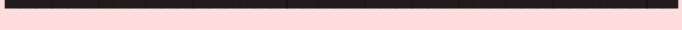
        magnetizations.append(abs(gs_mag))

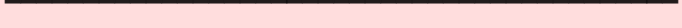
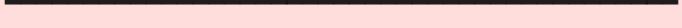
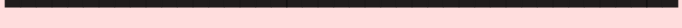
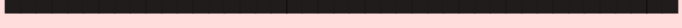
plt.plot(H_vals, magnetizations, label=f'{int(np.sqrt(N))}x{int(np.sqrt(N))}')

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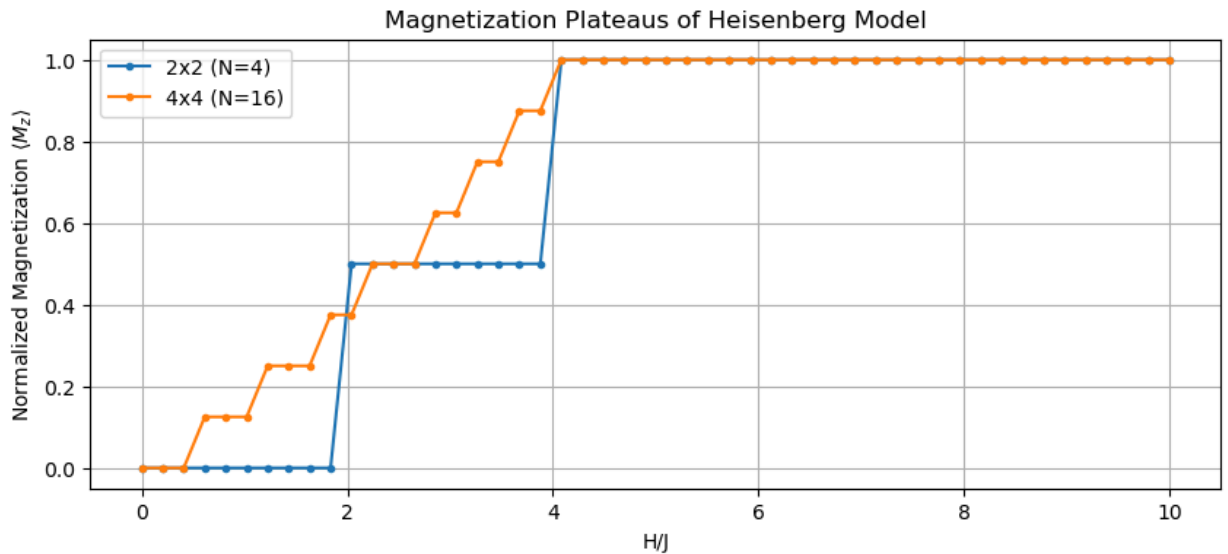
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plt.xlabel('H/J')  
plt.ylabel(r'Normalized Magnetization  $\langle M_z \rangle$ ')  
plt.title('Magnetization Plateaus of Heisenberg Model')  
plt.legend()  
plt.grid(True)  
plt.show()
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100%	17/17	[00:02<00:00,	7.23it/s]
100%	17/17	[00:02<00:00,	7.15it/s]
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```
In [2]: import numpy as np
import matplotlib.pyplot as plt

#params (dimensionless units of hbar = 1, m_B = 1, w_B = 1)
m_B, m_F = 1.0, 40.0/87.0 #like KRb mixture
w_B, w_F = 1.0, 1.0
g_B = 10.0
g_BF_vals = [0.0, 2.0, 15.0, -2.0] #non int, weak rep, phase sep, attractive
mu_B, mu_F = 15.0, 10.0 #chemical potentials fix the total particle numbers

r = np.linspace(0, 10, 500)
V_B = 0.5 * m_B * w_B**2 * r**2
V_F = 0.5 * m_F * w_F**2 * r**2

def solve_densities(g_BF, max_iter=500, mix=0.1):
    n_B = np.maximum(0, (mu_B - V_B) / g_B)
    n_F = np.zeros_like(r)

    for _ in range(max_iter):
        #LDA for fermions
        eff_mu_F = mu_F - V_F - g_BF * n_B
```

```

new_n_F = np.maximum(0, (1.0 / (6 * np.pi**2)) * (2 * m_F * np.maximum
#TF for bosons
eff_mu_B = mu_B - V_B - g_BF * new_n_F
new_n_B = np.maximum(0, eff_mu_B / g_B)

#mixing for numerical stabilization
n_B = mix * new_n_B + (1 - mix) * n_B
n_F = mix * new_n_F + (1 - mix) * n_F

return n_B, n_F

plt.figure(figsize=(12, 8))

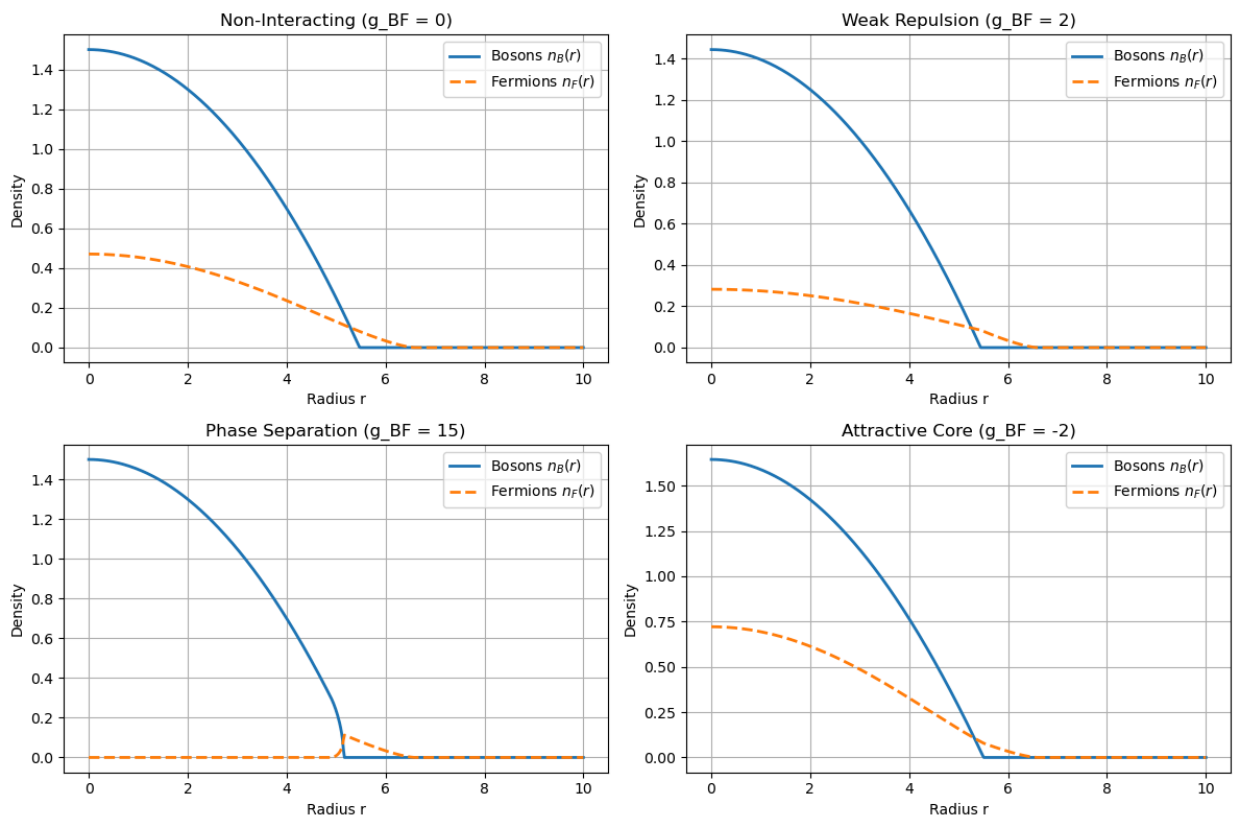
titles = ['Non-Interacting (g_BF = 0)', 'Weak Repulsion (g_BF = 2)',
          'Phase Separation (g_BF = 15)', 'Attractive Core (g_BF = -2)']

for i, g_BF in enumerate(g_BF_vals):
    n_B, n_F = solve_densities(g_BF)

    plt.subplot(2, 2, i+1)
    plt.plot(r, n_B, label='Bosons  $n_B(r)$ ', lw=2)
    plt.plot(r, n_F, label='Fermions  $n_F(r)$ ', lw=2, linestyle='--')
    plt.title(titles[i])
    plt.xlabel('Radius r')
    plt.ylabel('Density')
    plt.legend()
    plt.grid(True)

plt.tight_layout()
plt.show()

```



Running webapp

Putting above code I wrote into webapp via Streamlit with help of Gemini AI.

```
In [ ]: import streamlit as st
import numpy as np
import scipy.sparse as sp
import scipy.sparse.linalg as sla
import matplotlib.pyplot as plt

st.set_page_config(page_title="P452 Project 2: Many-Body Simulation", layout="")

st.title("P452 Project 2: Simulation of Many-Body System")
st.write("**Lecturer:** Cheng Chin | **Due:** May 7, 2026")

page = st.sidebar.radio("Navigation", ["Phase 1: Heisenberg Model", "Phase 2: I

# =====
# PHASE 1: HEISENBERG MODEL
# =====
if page == "Phase 1: Heisenberg Model":
    st.header("Phase 1: Exact Diagonalization of the Heisenberg Model")

    # --- Markdown Analysis for Phase 1 ---
    st.markdown(r"""
    ### Checkpoint 1.1: The 2-Site Dimer and Spin Frustration (Pen and Paper)

    **1. Hilbert Space**
    For  $N=2$  spins with  $S=1/2$ , each site has 2 states (up or down). The total
    The basis states are:  $|\uparrow\uparrow\rangle$ ,  $|\uparrow\downarrow\rangle$ ,  $|\downarrow\uparrow\rangle$ ,  $|\downarrow\downarrow\rangle$ 

    **2. Hamiltonian Matrix**
    The interaction is  $\hat{S}_1 \cdot \hat{S}_2 = \hat{S}_1^z \hat{S}_2^z + \frac{1}{2}(\hat{S}_1^+ \hat{S}_2^- + \hat{S}_1^- \hat{S}_2^+)$ 
    Let's apply the Hamiltonian to our basis states:
    *  $\hat{H}|\uparrow\uparrow\rangle = \left(\frac{J}{4} + H\right)|\uparrow\uparrow\rangle$ 
    *  $\hat{H}|\downarrow\downarrow\rangle = \left(\frac{J}{4} - H\right)|\downarrow\downarrow\rangle$ 
    *  $\hat{H}|\uparrow\downarrow\rangle = -\frac{J}{4}|\uparrow\downarrow\rangle + \frac{H}{2}(|\downarrow\uparrow\rangle - |\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$ 
    *  $\hat{H}|\downarrow\uparrow\rangle = -\frac{J}{4}|\downarrow\uparrow\rangle + \frac{H}{2}(|\uparrow\downarrow\rangle - |\downarrow\downarrow\rangle + |\uparrow\uparrow\rangle)$ 

    This gives the matrix in the ordered basis ( $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ )
    
$$H = \begin{pmatrix} \frac{J}{4} + H & 0 & 0 & 0 \\ 0 & \frac{J}{4} & H & 0 \\ 0 & H & \frac{J}{4} & 0 \\ 0 & 0 & 0 & \frac{J}{4} - H \end{pmatrix}$$


    **3. Ground State Transition**
    Diagonalizing the  $2 \times 2$  inner block yields the Singlet  $|S\rangle$  and Triplet states:
    *  $E_{T+} = \frac{J}{4} + H$  (from  $|\uparrow\uparrow\rangle$ )
    *  $E_{T-} = \frac{J}{4} - H$  (from  $|\downarrow\downarrow\rangle$ )
    *  $E_S = -\frac{3J}{4}$  (Singlet:  $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ )
    *  $E_{T0} = \frac{J}{4}$  (Triplet 0:  $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ )

    At  $H=0$ , the ground state is the Singlet ( $E = -3J/4$ ), representing an antiferromagnetic state.
    
$$\frac{J}{4} - |H_c| = -\frac{3J}{4} \implies |H_c| = J$$


    **4. Spin Frustration (3 Spins on a Ring)**
    * **Symmetry Reduction:** The total Hilbert space dimension is  $2^3 = 8$ .
    * **Calculation:** * **Sector  $+3/2$ ** ( $|\uparrow\uparrow\uparrow\rangle$ ):
```

```

* **Sector +1/2** (Basis:  $|\uparrow\uparrow\downarrow\downarrow\rangle$ ,  $|\uparrow\downarrow\uparrow\downarrow\rangle$ ,  $|\downarrow\uparrow\uparrow\downarrow\rangle$ ,  $|\downarrow\downarrow\uparrow\uparrow\rangle$ )
 $H_{1/2} = \begin{pmatrix} -\frac{J}{4} & \frac{H}{2} \\ \frac{H}{2} & \frac{J}{4} \end{pmatrix}$ 
The eigenvalues of this block are  $E = \frac{3J}{4} \pm \frac{H}{2}$  (noting that  $H = 2J \cos(\phi)$ )
* **Analysis:** At  $H=0$ , the ground state is in the  $\pm 1/2$  sectors with energy  $E = \pm J/4$ .

---
### Checkpoint 1.2: From 2x2 to 6x6 Square Lattices

**1. Hilbert Space and Dimension**
For a  $2 \times 2$  lattice,  $N=4$ . Total dimension =  $2^4 = 16$ .
For the  $S^z_{\text{total}} = 0$  sector (2 up, 2 down), the dimension is 6.
The basis states are:  $|\uparrow\uparrow\downarrow\downarrow\rangle, |\uparrow\downarrow\uparrow\downarrow\rangle, |\downarrow\uparrow\uparrow\downarrow\rangle, |\downarrow\downarrow\uparrow\uparrow\rangle, |\uparrow\downarrow\downarrow\uparrow\rangle, |\downarrow\uparrow\downarrow\uparrow\rangle$ .

**2. Role of the External Field**
The Zeeman term is  $\hat{H}_{\text{Zeeman}} = H \sum S^z_i$ . Since  $S^z_i = \pm 1/2$ , the Zeeman energy is  $\pm H/2$ .
Therefore,  $H \sum S^z_i |\phi\rangle = 0$ . The external field only shifts the energy levels.

**3. Eigenenergies and Limiting Regimes**
* ** $H=0$  or  $J=0$ :J=0, spins don't interact. At  $H=0$ ,  $J > 0$ , the ground state is degenerate.
* **General case  $H, J > 0$ :H increases, it tilts the energy of sectors with different  $S^z_{\text{total}}$ .

# --- Interactive Code UI for Phase 1 ---
st.markdown("----")
st.subheader("Interactive Numerical Simulation: Magnetization Staircase")

col1, col2 = st.columns(2)
with col1:
    N_selected = st.selectbox("Number of Spins (N)", [4, 16], index=0, help="N must be a power of 2")
    J_val = st.slider("Anti-ferromagnetic Coupling (J)", min_value=0.1, max_value=1.0, value=0.5)
with col2:
    H_max = st.slider("Max Magnetic Field (H)", min_value=1.0, max_value=2.0, value=1.5)
    num_points = st.slider("Resolution (Points)", min_value=10, max_value=100, value=50)

# Core Logic
@st.cache_data
def get_basis_sz(N, k):
    basis = []
    for i in range(2**N):
        if bin(i).count('1') == k:
            basis.append(i)
    return basis

@st.cache_data
def build_hamiltonian(N, J, H_field, Sz_target=None):
    L = int(np.sqrt(N))
    bonds = []
    for i in range(L):
        for j in range(L):
            idx = i * L + j
            right = i * L + (j + 1) % L
            down = ((i + 1) % L) * L + j
            bonds.extend([(idx, right), (idx, down)])

    if Sz_target is None:
        basis = list(range(2**N))
    else:
        k = int(N/2 + Sz_target)
        basis = get_basis_sz(N, k)

```

```

dim = len(basis)
state_to_idx = {state: i for i, state in enumerate(basis)}

rows, cols, data = [], [], []

for i, state in enumerate(basis):
    sz_total = sum(0.5 if (state & (1 << j)) else -0.5 for j in range(1, dim))
    rows.append(i); cols.append(i); data.append(H_field * sz_total)

    for s1, s2 in bonds:
        sz1 = 0.5 if (state & (1 << s1)) else -0.5
        sz2 = 0.5 if (state & (1 << s2)) else -0.5
        rows.append(i); cols.append(i); data.append(J * sz1 * sz2)

        if sz1 != sz2:
            new_state = state ^ ((1 << s1) | (1 << s2))
            if new_state in state_to_idx:
                j = state_to_idx[new_state]
                rows.append(i); cols.append(j); data.append(J / 2.0)

    return sp.csr_matrix((data, (rows, cols)), shape=(dim, dim))

if st.button("Run ED Simulation"):
    with st.spinner(f'Calculating Exact Diagonalization for N={N_selected}'):
        H_vals = np.linspace(0, H_max, num_points)
        magnetizations = []

        for H_field in H_vals:
            min_E = float('inf')
            gs_mag = 0
            for Sz in range(-N_selected//2, N_selected//2 + 1):
                H_mat = build_hamiltonian(N_selected, J_val, H_field, Sz_val)

                if H_mat.shape[0] <= 2:
                    evals = H_mat.toarray().diagonal()
                else:
                    evals = sla.eigsh(H_mat, k=1, which='SA', return_eigenvals=True)

                E0 = np.min(evals)
                if E0 < min_E:
                    min_E = E0
                    gs_mag = Sz / (N_selected/2)

            magnetizations.append(abs(gs_mag))

        fig, ax = plt.subplots(figsize=(8, 4))
        ax.plot(H_vals / J_val, magnetizations, label=f'{int(np.sqrt(N_selected))}')
        ax.set_xlabel('H / J')
        ax.set_ylabel(r'Normalized Magnetization $\langle M_z \rangle$')
        ax.set_title(f'Magnetization Plateaus (N={N_selected})')
        ax.legend()
        ax.grid(True)
        st.pyplot(fig)

# =====
# PHASE 2: BOSE-FERMI MIXTURES
# =====
elif page == "Phase 2: Bose-Fermi Mixtures":
    st.header("Phase 2: Hydrodynamics of Bose-Fermi Mixtures")

```

```

# --- Markdown Analysis for Phase 2 ---
st.markdown(r"""
### Checkpoint 2.1: Uniform Gas in the Ground State

**1. Bosonic Chemical Potential**
In the Thomas–Fermi approximation, we neglect the kinetic term  $-\frac{\hbar^2}{2m_B} \nabla^2 |\Psi_B|^2$ . Without fermions ( $a_{BF}=0$ ) and trap ( $V_B=0$ ), the GP equation reduces to

$$\mu_B = g_B n_B$$

Compressibility is  $\kappa \propto \left( \frac{\partial \mu_B}{\partial n_B} \right)^{-1}$ .

**2. Fermionic "Interaction"\mu_F = E_F = \frac{\hbar^2}{2m_F} (6\pi^2 n_F)^{2/3}. In the hydrodynamic equation, the term  $K_F(n_F)$  must equate to this chemical potential:

$$K_F(n_F) = \frac{\hbar^2}{2m_F} (6\pi^2 n_F)^{2/3}$$

**Scaling difference:** The bosonic mean-field term scales linearly with density, while the fermionic kinetic term scales as  $n_F^{2/3}$ .

**3. Interspecies Coupling**
With  $a_{BF} \neq 0$ , the uniform coupled equations become:

$$\mu_B = g_B n_B + g_{BF} n_F$$


$$\mu_F = E_F(n_F) + g_{BF} n_B$$


By taking  $n_B = (\mu_B - g_{BF} n_F) / g_B$  and substituting it into the fermion equation, we get:

$$\mu_F = E_F(n_F) + \frac{g_{BF} \mu_B}{g_B} - \frac{g_{BF}^2}{g_B} n_F$$

This creates an effective fermionic interaction  $g_{F,\text{eff}} = -\frac{g_{BF}^2}{g_B}$ .

**Stability Analysis**
The mixture is stable if  $\det(M) > 0$ :

$$M = \begin{pmatrix} g_B & g_{BF} \\ g_{BF} & \frac{\partial E_F}{\partial n_F} \end{pmatrix}$$


$$\det(M) = g_B \left( \frac{\hbar^2}{2m_F} (6\pi^2)^{2/3} n_F^{-1/3} \right) - g_{BF}^2 > 0$$


---
### Checkpoint 2.2: Mixtures in a Trap

**1 & 2. Thomas–Fermi and LDA Limits**
From the coupled equations including the harmonic trap  $V_i(r) = \frac{1}{2} m_i \omega_i^2 r^2$ :
* **Bosons:**  $n_B(r) = \max \left( 0, \frac{\mu_B - V_B(r) - g_{BF} n_F(r)}{g_B} \right)$ 
* **Fermions:**  $n_F(r) = \max \left( 0, \frac{1}{6\pi^2} \left[ \frac{2m_F}{\hbar^2} (\mu_F - V_F(r) - g_{BF} n_B(r)) \right]^{3/2} \right)$ 

**3. Physical Intuition**
* **Attractive ( $g_{BF} < 0$ ):** The  $-g_{BF} n_F(r)$  term increases the effective boson chemical potential, leading to higher density.
* **Repulsive ( $g_{BF} > 0$ ):** The species repel. If  $g_{BF}$  is large enough, it can lead to phase separation.

""")

# --- Interactive Code UI for Phase 2 ---
st.markdown("----")
st.subheader("Interactive Numerical Simulation: Coupled Hydrodynamic Density Profiles")

col1, col2, col3 = st.columns(3)
with col1:
    g_BF = st.slider("Interspecies Coupling (g_BF)", min_value=-5.0, max_value=5.0, value=0.0,
                    help="Negative = Attractive, Positive = Repulsive. High values lead to instability.")
    g_B = st.number_input("Boson Repulsion (g_B)", value=10.0)
with col2:
    m_B = st.number_input("Boson Mass (m_B)", value=1.0)
    m_F = st.number_input("Fermion Mass (m_F)", value=0.46) # e.g. K/Rb ratio
with col3:
    mu_B = st.number_input("Boson Chem. Potential (mu_B)", value=15.0)
    mu_F = st.number_input("Fermion Chem. Potential (mu_F)", value=10.0)

```



```

# Core Logic
def solve_densities(g_BF_val, max_iter=500, mix=0.1):
    r = np.linspace(0, 10, 500)
    w_B, w_F = 1.0, 1.0
    V_B = 0.5 * m_B * w_B**2 * r**2
    V_F = 0.5 * m_F * w_F**2 * r**2

    n_B = np.maximum(0, (mu_B - V_B) / g_B)
    n_F = np.zeros_like(r)

    for _ in range(max_iter):
        # LDA for Fermions
        eff_mu_F = mu_F - V_F - g_BF_val * n_B
        new_n_F = np.maximum(0, (1.0 / (6 * np.pi**2)) * (2 * m_F * np.max(
            eff_mu_F, 0)))

        # TF for Bosons
        eff_mu_B = mu_B - V_B - g_BF_val * new_n_F
        new_n_B = np.maximum(0, eff_mu_B / g_B)

        # Mixing for numerical stability (prevents oscillations in self-consistent)
        n_B = mix * new_n_B + (1 - mix) * n_B
        n_F = mix * new_n_F + (1 - mix) * n_F

    return r, n_B, n_F

r_vals, n_B_vals, n_F_vals = solve_densities(g_BF)

fig2, ax2 = plt.subplots(figsize=(10, 5))
ax2.plot(r_vals, n_B_vals, label='Bosons $n_B(r)$', lw=2, color='blue')
ax2.plot(r_vals, n_F_vals, label='Fermions $n_F(r)$', lw=2, linestyle='--')

if g_BF == 0:
    title_state = "Non-Interacting"
elif g_BF < 0:
    title_state = "Attractive Core"
elif g_BF > 10:
    title_state = "Phase Separation"
else:
    title_state = "Weak Repulsion"

ax2.set_title(f'Trap Density Profiles: {title_state} (g_BF = {g_BF})')
ax2.set_xlabel('Radius r')
ax2.set_ylabel('Density')
ax2.legend()
ax2.grid(True)
st.pyplot(fig2)

```

In []: # Auto save and launch P452 Project 2 Streamlit app

```

import os
import sys
import subprocess

# Full P452 Project 2 Streamlit Simulation (Heisenberg & Bose-Fermi)
# Notice the 'r' before the triple quotes below to make it a raw string!
streamlit_code = r'''import streamlit as st
import numpy as np
import scipy.sparse as sp
import scipy.sparse.linalg as sla
import matplotlib.pyplot as plt

```

```

# --- Page Configuration ---
st.set_page_config(page_title="P452 Project 2: Many-Body Simulation", layout="wide")

st.title("P452 Project 2: Simulation of Many-Body System")
st.write("**Lecturer:** Cheng Chin | **Due:** May 7, 2026")

# --- Sidebar Navigation ---
page = st.sidebar.radio("Navigation", ["Phase 1: Heisenberg Model", "Phase 2: Ising Model"])

# =====
# PHASE 1: HEISENBERG MODEL
# =====
if page == "Phase 1: Heisenberg Model":
    st.header("Phase 1: Exact Diagonalization of the Heisenberg Model")

    # --- Markdown Analysis for Phase 1 ---
    st.markdown(r"""
    ### Checkpoint 1.1: The 2-Site Dimer and Spin Frustration (Pen and Paper)

    **1. Hilbert Space**
    For  $N=2$  spins with  $S=1/2$ , each site has 2 states (up or down). The total Hilbert space dimension is  $2^N = 4$ .
    The basis states are:  $|\uparrow\uparrow\rangle$ ,  $|\uparrow\downarrow\rangle$ ,  $|\downarrow\uparrow\rangle$ ,  $|\downarrow\downarrow\rangle$ .

    **2. Hamiltonian Matrix**
    The interaction is  $\hat{S}_1 \cdot \hat{S}_2 = \hat{S}_1^x \hat{S}_2^x + \hat{S}_1^y \hat{S}_2^y + \hat{S}_1^z \hat{S}_2^z$ .
    Let's apply the Hamiltonian to our basis states:
    *  $\hat{H}|\uparrow\uparrow\rangle = \left(\frac{J}{4} + H\right)|\uparrow\uparrow\rangle$ 
    *  $\hat{H}|\downarrow\downarrow\rangle = \left(\frac{J}{4} - H\right)|\downarrow\downarrow\rangle$ 
    *  $\hat{H}|\uparrow\downarrow\rangle = -\frac{J}{4}|\uparrow\downarrow\rangle + H|\downarrow\uparrow\rangle$ 
    *  $\hat{H}|\downarrow\uparrow\rangle = -\frac{J}{4}|\downarrow\uparrow\rangle + H|\uparrow\downarrow\rangle$ 

    This gives the matrix in the ordered basis ( $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ ):
    
$$H = \begin{pmatrix} \frac{J}{4} + H & 0 & 0 & 0 \\ 0 & \frac{J}{4} & H & 0 \\ 0 & H & \frac{J}{4} & 0 \\ 0 & 0 & 0 & \frac{J}{4} - H \end{pmatrix}$$


    **3. Ground State Transition**
    Diagonalizing the  $2 \times 2$  inner block yields the Singlet  $|S\rangle$  and Triplet states.
    *  $E_{T+} = \frac{J}{4} + H$  (from  $|\uparrow\uparrow\rangle$ )
    *  $E_{T-} = \frac{J}{4} - H$  (from  $|\downarrow\downarrow\rangle$ )
    *  $E_S = -\frac{3J}{4}$  (Singlet:  $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ )
    *  $E_{T0} = \frac{J}{4}$  (Triplet 0:  $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ )

    At  $H=0$ , the ground state is the Singlet ( $E = -3J/4$ ), representing an antiferromagnetic state.
     $\frac{J}{4} - |H_c| = -\frac{3J}{4} \implies |H_c| = J$ 

    **4. Spin Frustration (3 Spins on a Ring)**
    * **Symmetry Reduction:** The total Hilbert space dimension is  $2^3 = 8$ .
    * **Calculation:** * **Sector  $+3/2$ ** ( $|\uparrow\uparrow\uparrow\rangle$ ):  $E = 3J/4 + H$ 
    * **Sector  $+1/2$ ** (Basis:  $|\uparrow\uparrow\downarrow\rangle, |\uparrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\rangle$ ):
    
$$H_{1/2} = \begin{pmatrix} -\frac{J}{4} + H & \frac{J}{2} & \frac{J}{2} \\ \frac{J}{2} & -\frac{J}{4} + H & \frac{J}{2} \\ \frac{J}{2} & \frac{J}{2} & -\frac{J}{4} + H \end{pmatrix}$$

    The eigenvalues of this block are  $E = \frac{3J}{4} + H$  (non-degenerate) and  $E = \frac{J}{4} + H$  (2-fold degenerate).
    * **Analysis:** At  $H=0$ , the ground state is in the  $\pm 1/2$  sectors with  $E = J/4$ .

    ---
    ### Checkpoint 1.2: From 2x2 to 6x6 Square Lattices

    **1. Hilbert Space and Dimension**
    For a  $2 \times 2$  lattice,  $N=4$ . Total dimension =  $2^4 = 16$ .
    For the  $S^z_{\text{total}} = 0$  sector (2 up, 2 down), the dimension is 6.
    The basis states are:  $|\uparrow\uparrow\downarrow\downarrow\rangle, |\uparrow\downarrow\uparrow\downarrow\rangle, |\uparrow\downarrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\downarrow\rangle, |\downarrow\uparrow\downarrow\uparrow\rangle, |\downarrow\downarrow\uparrow\uparrow\rangle$ .
    """
    )

```

2. Role of the External Field

The Zeeman term is $\hat{H}_{\text{Zeeman}} = H \sum \hat{S}_i^z$. Since $\langle \hat{S}_i^z \rangle = 0$ for all i , the Zeeman term only contributes to the energy of states with non-zero magnetization. Therefore, $\langle \hat{H}_{\text{Zeeman}} \rangle = H \langle \sum \hat{S}_i^z \rangle = 0$. The external field only shifts the energy levels.

3. Eigenenergies and Limiting Regimes

* ** $H=0$ or $J=0$:** At $J=0$, spins don't interact. At $H=0$, $J > 0$, the system is in a state of zero magnetization. * **General case $H, J > 0$:** As H increases, it tilts the energy of states with non-zero magnetization.

--- Interactive Code UI for Phase 1 ---

st.markdown("----")

st.subheader("Interactive Numerical Simulation: Magnetization Staircase")

col1, col2 = st.columns(2)

with col1:

N_selected = st.selectbox("Number of Spins (N)", [4, 16], index=0, help="Number of spins (N) must be even")

J_val = st.slider("Anti-ferromagnetic Coupling (J)", min_value=0.1, max_value=1.0, value=0.5)

with col2:

H_max = st.slider("Max Magnetic Field (H)", min_value=1.0, max_value=2.0, value=1.5)

num_points = st.slider("Resolution (Points)", min_value=10, max_value=100, value=50)

Core Logic

@st.cache_data

def get_basis_sz(N, k):

basis = []

for i in range(2*N):

if bin(i).count('1') == k:

basis.append(i)

return basis

@st.cache_data

def build_hamiltonian(N, J, H_field, Sz_target=None):

L = int(np.sqrt(N))

bonds = []

for i in range(L):

for j in range(L):

idx = i * L + j

right = i * L + (j + 1) % L

down = ((i + 1) % L) * L + j

bonds.extend([(idx, right), (idx, down)])

if Sz_target is None:

basis = list(range(2*N))

else:

k = int(N/2 + Sz_target)

basis = get_basis_sz(N, k)

dim = len(basis)

state_to_idx = {state: i for i, state in enumerate(basis)}

rows, cols, data = [], [], []

for i, state in enumerate(basis):

sz_total = sum(0.5 if (state & (1 << j)) else -0.5 for j in range(L))
rows.append(i); cols.append(i); data.append(H_field * sz_total)

for s1, s2 in bonds:

sz1 = 0.5 if (state & (1 << s1)) else -0.5

sz2 = 0.5 if (state & (1 << s2)) else -0.5

```

        rows.append(i); cols.append(i); data.append(J * sz1 * sz2)

    if sz1 != sz2:
        new_state = state ^ ((1 << s1) | (1 << s2))
        if new_state in state_to_idx:
            j = state_to_idx[new_state]
            rows.append(i); cols.append(j); data.append(J / 2.0)

    return sp.csr_matrix((data, (rows, cols)), shape=(dim, dim))

if st.button("Run ED Simulation"):
    with st.spinner(f"Calculating Exact Diagonalization for N={N_selected}"):
        H_vals = np.linspace(0, H_max, num_points)
        magnetizations = []

        for H_field in H_vals:
            min_E = float('inf')
            gs_mag = 0
            for Sz in range(-N_selected//2, N_selected//2 + 1):
                H_mat = build_hamiltonian(N_selected, J_val, H_field, Sz_to_idx)

                if H_mat.shape[0] <= 2:
                    evals = H_mat.toarray().diagonal()
                else:
                    evals = sla.eigsh(H_mat, k=1, which='SA', return_eigenvals=True)

                E0 = np.min(evals)
                if E0 < min_E:
                    min_E = E0
                    gs_mag = Sz / (N_selected/2)

            magnetizations.append(abs(gs_mag))

        fig, ax = plt.subplots(figsize=(8, 4))
        ax.plot(H_vals / J_val, magnetizations, label=f"{int(np.sqrt(N_selected))}")
        ax.set_xlabel('H / J')
        ax.set_ylabel(r'Normalized Magnetization $\langle M_z \rangle$')
        ax.set_title(f'Magnetization Plateaus (N={N_selected})')
        ax.legend()
        ax.grid(True)
        st.pyplot(fig)

# =====
# PHASE 2: BOSE-FERMI MIXTURES
# =====
elif page == "Phase 2: Bose-Fermi Mixtures":
    st.header("Phase 2: Hydrodynamics of Bose-Fermi Mixtures")

    # --- Markdown Analysis for Phase 2 ---
    st.markdown(r"""
    ### Checkpoint 2.1: Uniform Gas in the Ground State

    **1. Bosonic Chemical Potential**
    In the Thomas-Fermi approximation, we neglect the kinetic term  $-\frac{\hbar^2}{2m} \nabla^2 \Psi_B$ .
    Without fermions ( $a_{BF}=0$ ) and trap ( $V_B=0$ ), the GP equation reduces to
    
$$\mu_B = g_B |\Psi_B|^2 = g_B n_B$$

    Compressibility is  $\kappa \propto \left( \frac{\partial \mu_B}{\partial n_B} \right)^{-1}$ 

    **2. Fermionic "Interaction"

```

For a uniform Fermi gas, $\mu_F = E_F = \frac{\hbar^2}{2m_F}(6\pi^2 n_F)^{2/3}$. In the hydrodynamic equation, the term $K_F(n_F)$ must equate to this chemical potential μ_F .

$$K_F(n_F) = \frac{\hbar^2}{2m_F}(6\pi^2 n_F)^{2/3}$$

****Scaling difference:**** The bosonic mean-field term scales linearly with density, while the fermionic term scales with $n_F^{2/3}$.

****3. Interspecies Coupling****
 With $a_{BF} \neq 0$, the uniform coupled equations become:

$$\mu_B = g_B n_B + g_{BF} n_F$$

$$\mu_F = E_F(n_F) + g_{BF} n_B$$

By taking $n_B = (\mu_B - g_{BF} n_F) / g_B$ and substituting it into the fermion equation, we get:

$$\mu_F = E_F(n_F) + \frac{g_{BF}}{g_B} \mu_B - \frac{g_{BF}^2}{g_B} n_F$$
 This creates an effective fermionic interaction $g_{F,\text{eff}} = -\frac{g_{BF}^2}{g_B}$.

****Stability Analysis:****
 The mixture is stable if $\det(M) > 0$:

$$M = \begin{pmatrix} g_B & g_{BF} \\ g_{BF} & \frac{\partial E_F}{\partial n_F} \end{pmatrix}$$

$$\det(M) = g_B \left(\frac{\hbar^2}{3m_F}(6\pi^2)^{2/3} n_F^{-1/3} \right) - g_{BF}^2 > 0$$

Checkpoint 2.2: Mixtures in a Trap

****1 & 2. Thomas-Fermi and LDA Limits****
 From the coupled equations including the harmonic trap $V_i(r) = \frac{1}{2} m_i \omega^2 r^2$, we solve for the densities $n_B(r)$ and $n_F(r)$.
****Bosons:**** $n_B(r) = \max(0, \frac{\mu_B - V_B(r) - g_{BF} n_F(r)}{g_B})$
****Fermions:**** $n_F(r) = \max(0, \frac{1}{6\pi^2} \left[\frac{2m_F(\mu_F - V_F(r) + g_{BF} n_B(r))}{\hbar^2} \right]^{3/2})$

****3. Physical Intuition****
****Attractive ($g_{BF} < 0$):**** The $-g_{BF} n_F(r)$ term increases the effective boson chemical potential, leading to higher boson density and lower fermion density.
****Repulsive ($g_{BF} > 0$):**** The species repel. If g_{BF} is large enough, the bosons can form a condensate even at finite temperatures.

```
# --- Interactive Code UI for Phase 2 ---
st.markdown("----")
st.subheader("Interactive Numerical Simulation: Coupled Hydrodynamic Densities")

col1, col2, col3 = st.columns(3)
with col1:
    g_BF = st.slider("Interspecies Coupling (g_BF)", min_value=-5.0, max_value=5.0, value=0.0,
                    help="Negative = Attractive, Positive = Repulsive. High values increase repulsion.")
    g_B = st.number_input("Boson Repulsion (g_B)", value=10.0)
with col2:
    m_B = st.number_input("Boson Mass (m_B)", value=1.0)
    m_F = st.number_input("Fermion Mass (m_F)", value=0.46) # e.g. K/Rb ratio
with col3:
    mu_B = st.number_input("Boson Chem. Potential (mu_B)", value=15.0)
    mu_F = st.number_input("Fermion Chem. Potential (mu_F)", value=10.0)

# Core Logic
def solve_densities(g_BF_val, max_iter=500, mix=0.1):
    r = np.linspace(0, 10, 500)
    w_B, w_F = 1.0, 1.0
    V_B = 0.5 * m_B * w_B**2 * r**2
    V_F = 0.5 * m_F * w_F**2 * r**2

    n_B = np.maximum(0, (mu_B - V_B) / g_B)
    n_F = np.zeros_like(r)

    for _ in range(max_iter):
        # LDA for Fermions
```

```

    eff_mu_F = mu_F - V_F - g_BF_val * n_B
    new_n_F = np.maximum(0, (1.0 / (6 * np.pi**2)) * (2 * m_F * np.max(

    # TF for Bosons
    eff_mu_B = mu_B - V_B - g_BF_val * new_n_F
    new_n_B = np.maximum(0, eff_mu_B / g_B)

    # Mixing for numerical stability (prevents oscillations in self-co
    n_B = mix * new_n_B + (1 - mix) * n_B
    n_F = mix * new_n_F + (1 - mix) * n_F

    return r, n_B, n_F

r_vals, n_B_vals, n_F_vals = solve_densities(g_BF)

fig2, ax2 = plt.subplots(figsize=(10, 5))
ax2.plot(r_vals, n_B_vals, label='Bosons $n_B(r)$', lw=2, color='blue')
ax2.plot(r_vals, n_F_vals, label='Fermions $n_F(r)$', lw=2, linestyle='--')

if g_BF == 0:
    title_state = "Non-Interacting"
elif g_BF < 0:
    title_state = "Attractive Core"
elif g_BF > 10:
    title_state = "Phase Separation"
else:
    title_state = "Weak Repulsion"

ax2.set_title(f'Trap Density Profiles: {title_state} (g_BF = {g_BF})')
ax2.set_xlabel('Radius r')
ax2.set_ylabel('Density')
ax2.legend()
ax2.grid(True)
st.pyplot(fig2)
...

# Ensure the directory exists
folder = "/Users/pouyasadeghpour/Desktop/Desktop/PhD/UChicago/Quantum_Optics_G
filename = "streamlit_app_p452_proj2.py"
filepath = os.path.join(folder, filename)

# Create the directory if it doesn't exist to prevent file write errors
os.makedirs(folder, exist_ok=True)

with open(filepath, "w") as f:
    f.write(streamlit_code)

print(f"Streamlit app saved to: {filepath}")

print("To run your app manually, open terminal and execute:")
print(f"  streamlit run {filepath}\n")

# Auto-launch the app
try:
    subprocess.Popen([sys.executable, "-m", "streamlit", "run", filepath])
    print("Streamlit app launching...")
except Exception as e:
    print("Failed to launch Streamlit automatically.")
    print(f"Error: {e}")

```

In []: